

AI chatbots with shinychat (**R**) :: CHEAT SHEET



Create a basic chatbot

1. Connect to LLM

Choose model provider and provide instructions

2. Add capabilities

Register tools like web search

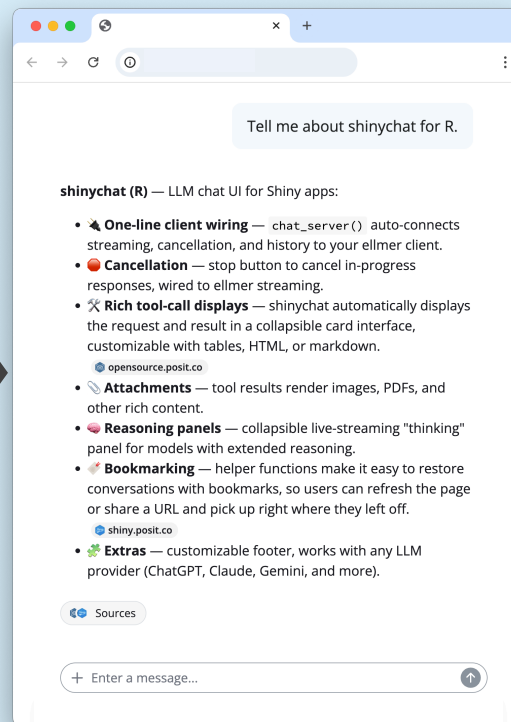
3. Create front-end

Create and display the Chat .ui()

```
# app.R
library(shiny)
library(shinychat)

client <- ellmer::chat_anthropic(
  model = "claude-sonnet-5",
  system_prompt = "You are a helpful assistant."
)
client$register_tool(ellmer::claude_tool_web_search())
client$register_tool(ellmer::claude_tool_web_fetch())

ui <- bslib::page_fillable(
  chat_ui("chat", greeting = "Welcome to my chat!")
)
server <- function(input, output, session) {
  chat_server("chat", client)
}
shinyApp(ui, server)
```



Chat features

BUILT-IN

With no extra effort, get the following

- **History:** archive, revisit, and share conversations.
- **Attachments:** include images, pdfs, plain text, & more.
- **Cancel & edit:** input the wrong prompt? Cancel, revise and resubmit.
- **Tool displays:** let users inspect, validate, and learn from code the model executes.
- **Thinking display:** inspect how LLMs reason through hard problems.
- **Rich streaming output:** performantly stream LLM (markdown) responses with rich formatting for code blocks, images, and more.

OPT-IN

With extra effort, add the following

- **Greetings:** supply a useful intro to the chat
- **Suggestions:** make it easy to get started
- **Slash commands:** useful shortcuts for common workflows

Connect to LLM

CHOOSE A PROVIDER

Initialize an LLM chat provider via `ellmer`.

Anthropic Claude: `chat_claude()`
OpenAI: `chat_openai()`
Google Gemini: `chat_google()`
AWS Bedrock: `chat_aws_bedrock()`
Azure OpenAI: `chat_azure_openai()`
Databricks: `chat_databricks()`
Posit AI: `chat_posit()`
LMStudio: `chat_lmstudio()`

Many require an API key.
Provide yours to
`edit_r_environ()` from
`usethis`.

Visit <https://ellmer.tidyverse.org/reference/index.html> to see **all available** providers and their **prerequisites**.

QUICKLY ITERATE

Launch a chat app with one line

```
client = chat_claude()
live_browser(client)
```

STREAM OUTPUT ANYWHERE

Not only in Shiny, but at the console, etc.

```
client = chat_claude()
client$chat("Hello!")
```

First impression

GREET & SUGGEST

Customize what users see on first page load.

Supply markdown/HTML for a custom greeting. Greetings can include special HTML suggestion markup.

```
chat_ui(greeting="""
### 🙌 Welcome!
```

```
Not sure what to do? Try:
1. <span class="suggestion">Tell me a
joke</span>
2. <span class="suggestion">Write a
poem</span>
""")
```

🙌 **Welcome!**

I'm your Shiny app-building assistant — ask me about layouts, reactivity, styling, or debugging.

Try one of these:

Add a sidebar with filters →

Why isn't my plot updating? →

Layouts

LOCATE THE CHAT IN A LARGER SHINY APP

Use shiny/bslib to customize layout and functionality

SCREEN-FILLING LAYOUT

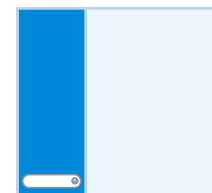


Make contents (chat) **fill page**

```
ui <- page_fillable(
  chat_ui("chat"),
  fillable_mobile = TRUE
)
```

Also fill on mobile

SIDEBAR LAYOUT



```
ui <- page_sidebar(
  sidebar = sidebar(
    chat_ui("chat"),
    fillable = TRUE
  )
)
```

Make contents (chat) **fill sidebar**

CARD LAYOUT



```
ui <- page_fillable(
  card(
    chat_ui(id = "chat")
  )
)
```

Make contents (card) **fill page**

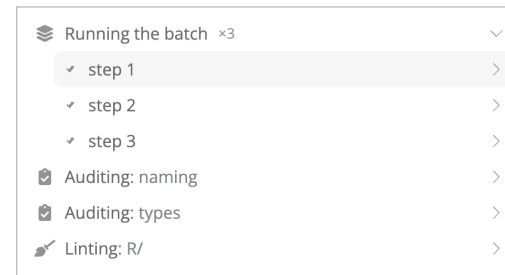
Cards are fillable by default

Extras

Tool Displays

Inform users of actions taken by the LLM

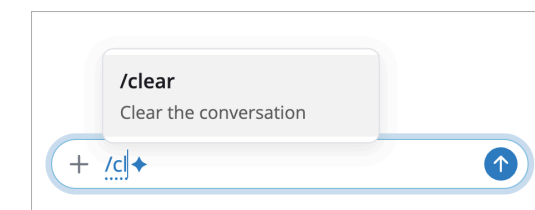
Tool call information is automatically displayed to the user (for transparency). The display is also fully customizable.



Slash commands

Add shortcuts for common workflows

```
chat$slash_command("clear", "Clear", \(){
  chat$clear()
})
```



AI chatbots with shinychat (**Python**) : : CHEAT SHEET



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3. Create front-end

Create and display the Chat .ui()

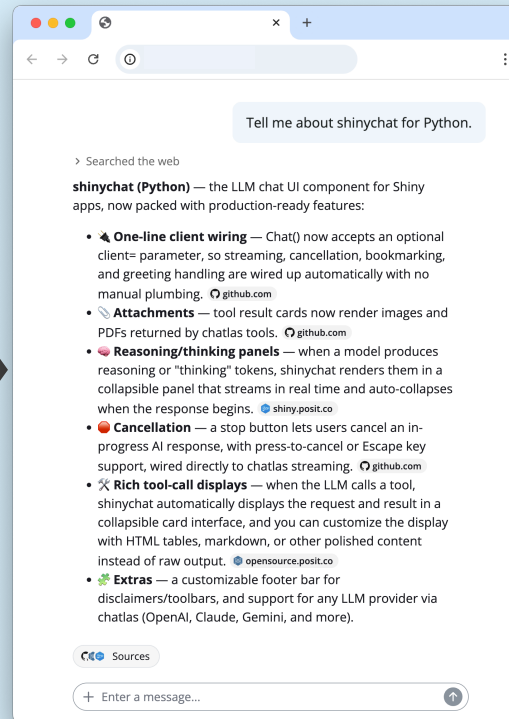
```
# app.py
import chatlas as ctl
from shinychat.express import Chat

client = ctl.ChatAnthropic(
    model="claude-sonnet-5",
    system_prompt="You are a helpful assistant."
)

client.register_tool(ctl.tool_web_search())
client.register_tool(ctl.tool_web_fetch())

chat = Chat(id="chat", client=client)
chat.ui(greeting="Welcome to my custom chat!")
```

This cheatsheet uses **Shiny Express**. To see examples for Shiny Core, see <https://shiny.posit.co/py/docs/genai-chatbots.html>.



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OpenAI: **ChatOpenAI()**

Google Gemini: **ChatGoogle()**

AWS Bedrock: **ChatBedrock()**

Azure OpenAI: **ChatAzureOpenAI()**

Databricks: **ChatDatabricks()**

Posit AI: **ChatPosit()**

LMStudio: **ChatLMStudio()**

Many require an API key. Place yours in `.env` file and load with `dotenv`.

Visit <https://posit-dev.github.io/chatlas/reference> to see **all available** providers and their **prerequisites**.

QUICKLY ITERATE

Launch a chat app with one line

```
client = ctl.ChatAnthropic()
client.app()
```

STREAM OUTPUT ANYWHERE

Not only in Shiny, but in notebooks, the console, etc.

```
client = ctl.ChatAnthropic()
client.chat("Hello!")
```

First impression

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Customize what users see on first page load.

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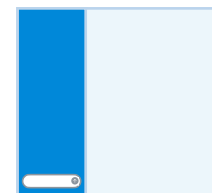


Make contents (Chat) fill page

```
ui.page_opts(
  fillable=True,
  fillable_mobile=True,
)

chat = ui.Chat(id="chat")
chat.ui()
```

SIDEBAR LAYOUT



Make contents (chat) fill sidebar

```
chat = ui.Chat(id="chat")

with ui.sidebar(fillable=True):
  chat.ui()
```

"Main content"

CARD LAYOUT



Make contents (card) fill page

Cards are fillable by default

```
ui.page_opts(
  fillable=True,
  fillable_mobile=True
)

chat = ui.Chat(id="chat")

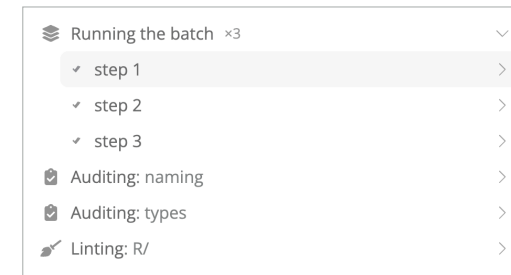
with ui.card():
  chat.ui()
```

Extras

Tool Displays

Inform users of actions taken by the LLM

Tool call information is automatically displayed to the user (for transparency). The display is also fully customizable.



Slash commands

Add shortcuts for common workflows

```
@chat.slash_command("clear", "Clear")
async def _():
    await chat.clear()
```

